

Tarun Mittal

Senior Software Engineer

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SUMMARY

- Senior Software Engineer with **5 years of experience** designing and owning cloud-native, high-throughput distributed systems.
- Expertise in **Java/Spring Boot**, Hibernate/JPA, **ClickHouse/PostgreSQL**, and Python for Agentic AI/RAG.
- Led backend migrations end to end and **mentored engineers** at Uber, Masters India, and Impact Analytics.

SKILLS

- **Languages:** Java, SQL, Python, C, C++.
- **Backend & APIs:** Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST, gRPC, Spring Batch, Design Patterns.
- **AI & Agents:** Python (FastAPI, LangGraph, LangChain), LLM agents (OpenAI/Gemini/Claude), MCP, RAG, text-to-SQL, pgvector.
- **Streaming & Data:** Apache Kafka, Apache Flink, Apache Spark, Apache Pinot, ClickHouse, BigQuery, CDC, CQRS.
- **Databases & Storage:** PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, Amazon S3, Caching.
- **Cloud & Infrastructure:** GCP, AWS, Docker, Kubernetes, CI/CD, Maven, Bazel, ELK, Grafana, New Relic.
- **Core Engineering:** Distributed Systems, HLD/LLD, Concurrency, High Availability, Mentorship, JUnit, Mockito.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

Agentic AssortSmart Platform & PostgreSQL to ClickHouse Data Migration

- Built a **Python LangGraph/MCP clustering copilot** with 14 audited tools; targets: days to under 1h, 1 to 20+ configurations, 8.5% to under 2% failures.
- Benchmarked a **23.7M-row** Order Batching query at **3.86s on ClickHouse vs 3m 40s+ on PostgreSQL**, a **60x** improvement.
- Sustained **5.9M vs 250K rows/sec** on bulk loads and improved query-to-CSV export throughput **43x**, from 607 to 26.4K rows/sec.
- Designed a **CQRS/CDC migration**: PostgreSQL writes, ClickHouse reads, daily dimension refresh, and **Redis RYW routing with a 30s TTL**.
- Cut 10K-row updates on a **29M-row fact table from 39s to 7s** via partition merges; compressed 17M rows **7.8x** to 433 MiB.
- **Tech Used:** Python, LangGraph, MCP, ClickHouse, PostgreSQL, CDC, Redis, Kafka, GCP, Docker.

Software Development Engineer 2 UBER (via EPAM Systems), Bangalore July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Built Uber's **FRM Risk Scoping platform** across **8 screens and 30+ APIs**, targeting 70% less manual work at **\$340M materiality**.
- Designed an **11-table MySQL schema** using Spring Data JPA/Hibernate and controller, service, repository layers with header-based auth.
- Migrated Sheets-backed reconciliation to **MySQL v2 APIs**, validated HFM against 10-Q data, and fixed a column-aliasing bug.
- **Led 3 engineers** and grew the service to **1,100+ unit tests** across controllers, services, and repositories.
- **Tech Used:** Java, Spring Boot, Spring Data JPA, Hibernate, MySQL, Redis, JUnit, Docker.

Menu Ingestion & Automation Platform (Uber Eats)

- Built a **Kafka/Flink/Spark** pipeline for 30K+ menus/month; cut onboarding 24h to 2h and saved **\$600K+/year**.
- Built a **Python RAG/Gemini** pipeline for PDFs and images, reaching **98% fidelity and 100% schema consistency** in offline evaluation.
- Raised ingestion success about 60% to 95%+ via proxies and IP rotation; **Pinot** dashboards cut detection from hours to minutes.
- **Tech Used:** Python, Apache Kafka, Apache Flink, Apache Spark, Apache Pinot, Selenium, Gemini, GCP, Docker.

Software Development Engineer 2

MASTERS INDIA, Noida

December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Led a PHP-to-**Spring Boot microservices** migration and mentored **2 engineers**, cutting p95 latency **1.2s to 300ms** for 1,500+ clients.
- Scaled **Spring Batch** imports to 100K+ transactions and 1M+ daily, raising sustained throughput **700 to 4,000 requests/min**.
- Added **Redis cache-aside** to cut redundant DB reads 30%; shipped audit logs that reduced client churn **15%**.
- Added **ELK/New Relic** alerting, cutting triage 70%; raised coverage **35% to 82%** with a 98% deployment success rate.
- **Tech Used:** Java, Spring Boot, Hibernate, PostgreSQL, MongoDB, Redis, Elasticsearch, Spring Batch, Docker, ELK, New Relic, AWS.

Software Development Engineer

GEEKSFORGEEKS, Noida

August 2021 – November 2022

- Migrated PHP services to **Spring Boot**, improving reliability for **10K+ daily queries**.
- Built voting and pinning APIs using MySQL, MongoDB, Redis, and Elasticsearch, lifting **premium subscriptions 15 to 20%**.
- Built influencer analytics and scheduled jobs for video processing, raising **course sales 30%** and operational efficiency 70%.

EDUCATION

B.Tech (Information Technology), CGPA 7.7/10

Institute of Engineering & Technology,
Lucknow

July 2017 – June 2021

CERTIFICATIONS

- [HackerRank Problem Solving Certificate](#).
- [LangChain Academy Certificate](#).

ACHIEVEMENTS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier Round 2021.
- Finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.